

Do Carmo §1.2 Problem Set

Exercise 1. Find a parametrized curve $\alpha(t)$ whose trace is the circle $x^2 + y^2 = 1$ such that $\alpha(t)$ runs clockwise around the circle with $\alpha(0) = (0, 1)$.

Solution. The standard counterclockwise parametrization of a unit circle starting at $(1, 0)$ is $(\cos t, \sin t)$. To start at $(0, 1)$ and move clockwise, the x -coordinate must increase into positive values while the y -coordinate must decrease from 1 as t increases from 0.

Using the trigonometric identities for a shift and reflection, we obtain:

$$\alpha(t) = (\sin t, \cos t, 0)$$

Let us verify the given boundary conditions:

- (1) **Initial Position:** $\alpha(0) = (\sin 0, \cos 0, 0) = (0, 1, 0)$, which satisfies the condition.
- (2) **Trace Constraint:** $x^2 + y^2 = \sin^2 t + \cos^2 t = 1$, confirming the curve lies on the unit circle.
- (3) **Direction:** At a small time step $t > 0$, $\sin t$ becomes positive and $\cos t$ decreases below 1, moving the point into the first quadrant toward $(1, 0)$, which is a clockwise rotation.

□

Exercise 2. Let $\alpha(t)$ be a parametrized curve which does not pass through the origin. If $\alpha(t_0)$ is the point of the trace of α closest to the origin and $\alpha'(t_0) \neq 0$, show that the position vector $\alpha(t_0)$ is orthogonal to $\alpha'(t_0)$.

Solution. Let $d(t) = |\alpha(t)|$ represent the distance from the curve to the origin at any time t . Minimizing the distance function is equivalent to minimizing the squared distance function $f(t) = |\alpha(t)|^2$, which avoids square roots. We express this using the vector dot product:

$$f(t) = \alpha(t) \cdot \alpha(t)$$

We are given that the curve reaches its closest point to the origin at $t = t_0$. Therefore, t_0 is a local minimum of the differentiable function $f(t)$. By Fermat's theorem on stationary points, the first derivative must vanish at this point:

$$f'(t_0) = 0$$

Applying the product rule for dot products to differentiate $f(t)$, we find:

$$f'(t) = \frac{d}{dt}(\alpha(t) \cdot \alpha(t)) = \alpha'(t) \cdot \alpha(t) + \alpha(t) \cdot \alpha'(t) = 2\alpha(t) \cdot \alpha'(t)$$

Evaluating this derivative at the local minimum $t = t_0$:

$$2\alpha(t_0) \cdot \alpha'(t_0) = 0 \implies \alpha(t_0) \cdot \alpha'(t_0) = 0$$

Since their dot product is zero, the position vector $\alpha(t_0)$ is orthogonal to the velocity vector $\alpha'(t_0)$. □

Exercise 3. A parametrized curve $\alpha(t)$ has the property that its second derivative $\alpha''(t)$ is identically zero. What can be said about α ?

Solution. We are given that $\alpha''(t) = (0, 0, 0)$ for all t in the domain. We can determine the nature of $\alpha(t)$ by integrating this vector differential equation component by component.

Integrating the second derivative vector once with respect to t yields a constant velocity vector:

$$\alpha'(t) = \int \alpha''(t) dt = \mathbf{v}$$

where $\mathbf{v} = (v_1, v_2, v_3)$ is a fixed constant vector in $\mathbb{R}^3$. Integrating a second time gives the position vector function:

$$\alpha(t) = \int \alpha'(t) dt = \mathbf{v}t + \mathbf{p}$$

where $\mathbf{p} = (p_1, p_2, p_3)$ is another constant integration vector representing the initial position $\alpha(0)$.

Depending on the value of the constant vector $\mathbf{v}$, two geometric cases arise:

Case 1: If $\mathbf{v} \neq \mathbf{0}$, the curve $\alpha(t) = \mathbf{p} + t\mathbf{v}$ is a **straight line** passing through $\mathbf{p}$ parallel to $\mathbf{v}$.

Case 2: If $\mathbf{v} = \mathbf{0}$, the curve collapses to a single **stationary point** $\alpha(t) = \mathbf{p}$ for all t .

□

Exercise 4. Let $\alpha : I \rightarrow \mathbb{R}^3$ be a parametrized curve and let $v \in \mathbb{R}^3$ be a fixed vector. Assume that $\alpha'(t)$ is orthogonal to v for all $t \in I$ and that $\alpha(0)$ is also orthogonal to v . Prove that $\alpha(t)$ is orthogonal to v for all $t \in I$.

Solution. To show that $\alpha(t)$ remains orthogonal to the fixed vector v for all time, we define a scalar real-valued function $h(t)$ that tracks their dot product:

$$h(t) = \alpha(t) \cdot v$$

We differentiate $h(t)$ with respect to t . Since v is a constant vector ($\frac{dv}{dt} = \mathbf{0}$), the product rule yields:

$$h'(t) = \alpha'(t) \cdot v + \alpha(t) \cdot \frac{dv}{dt} = \alpha'(t) \cdot v$$

We are given that the velocity vector $\alpha'(t)$ is orthogonal to v for all $t \in I$, meaning $\alpha'(t) \cdot v = 0$. Substituting this into our derivative expression gives:

$$h'(t) = 0 \quad \text{for all } t \in I$$

Because its derivative is identically zero on the interval, $h(t)$ must be a constant value C :

$$h(t) = C$$

To find the value of C , we evaluate the function at the initial boundary condition $t = 0$. We are given that $\alpha(0)$ is orthogonal to v , so:

$$C = h(0) = \alpha(0) \cdot v = 0$$

Consequently, $h(t) = 0$ for all $t \in I$, which proves that $\alpha(t) \cdot v = 0$, meaning $\alpha(t)$ is orthogonal to v for all $t \in I$. $\square$

Exercise 5. Let $\alpha : I \rightarrow \mathbb{R}^3$ be a parametrized curve, with $\alpha'(t) \neq 0$ for all $t \in I$. Show that $|\alpha(t)|$ is a nonzero constant if and only if $\alpha(t)$ is orthogonal to $\alpha'(t)$ for all $t \in I$.

Solution. Let us define the auxiliary squared magnitude function $g(t) = |\alpha(t)|^2 = \alpha(t) \cdot \alpha(t)$. Differentiating this function gives $g'(t) = 2\alpha(t) \cdot \alpha'(t)$. We prove the bi-conditional statement in two directions:

($\implies$) Assume $|\alpha(t)| = c$ where c is a nonzero constant. Then the squared function is also constant:

$$g(t) = c^2$$

Taking the derivative of a constant yields zero:

$$g'(t) = 0 \implies 2\alpha(t) \cdot \alpha'(t) = 0 \implies \alpha(t) \cdot \alpha'(t) = 0$$

This proves that the position vector $\alpha(t)$ is orthogonal to the tangent vector $\alpha'(t)$ for all $t \in I$.

($\impliedby$) Assume $\alpha(t)$ is orthogonal to $\alpha'(t)$ for all $t \in I$. By definition of orthogonality, their dot product vanishes:

$$\alpha(t) \cdot \alpha'(t) = 0$$

This directly implies that the derivative of our squared magnitude function is zero:

$$g'(t) = 2\alpha(t) \cdot \alpha'(t) = 0$$

Since its derivative is zero across the interval, $g(t)$ must equal a constant value C :

$$\alpha(t) \cdot \alpha(t) = C$$

Taking the square root gives $|\alpha(t)| = \sqrt{C}$. We must show that $C > 0$ so that the constant is nonzero. If $C = 0$, then $\alpha(t) = \mathbf{0}$ for all $t \in I$. However, the derivative of a constant zero vector is $\alpha'(t) = \mathbf{0}$, which directly contradicts the problem's explicit condition that $\alpha'(t) \neq 0$ everywhere. Thus, C must be strictly positive, meaning $|\alpha(t)| = \sqrt{C}$ is a nonzero constant. $\square$